

Outline

Recap: overfitting

- ▶ Overfitting: model fits noise / artifacts in the training data
- ▶ Symptom: small *training* error, large *test* error
- ▶ L02 example: degree-15 polynomial on a tiny dataset
- ▶ Three high-level fixes:
 1. Collect more data (often infeasible)
 2. Reduce hypothesis space (binary feature in/out — coarse)
 3. Penalize complexity *continuously* $\Rightarrow$ **regularization**



What is regularization?

Methods that inject an **inductive bias** (typically “low complexity”) into the model to reduce overfitting and improve the bias-variance tradeoff.

- ▶ **Explicit** regularization: add a complexity penalty to ERM (L_1 , L_2).
- ▶ **Implicit** regularization: early stopping, data augmentation, parameter sharing, dropout, ensembling.
- ▶ **Structured** regularization: prior knowledge over groups of parameters (e.g., group lasso).

This lecture focuses on **explicit** L_1 and L_2 penalties on the coefficients.

Regularized empirical risk minimization

Trade off two competing goals:

- ▶ Maximize **fit** (minimize training loss)
- ▶ Minimize **complexity** of the model

Combine into a single objective:

$$\mathcal{R}_{\text{reg}}(\boldsymbol{\theta}) = \mathcal{R}_{\text{emp}}(\boldsymbol{\theta}) + \lambda \cdot J(\boldsymbol{\theta})$$

- ▶ $J(\boldsymbol{\theta})$: **complexity penalty** / regularizer
- ▶ $\lambda \geq 0$: **regularization strength**
- ▶ $\lambda = 0$: plain ERM
- ▶ $\lambda \rightarrow \infty$: model becomes as “simple” as possible (often 0 coefficients)
- ▶ Choosing λ is a tuning problem — usually via cross-validation (L05)

Bias and variance, intuitively

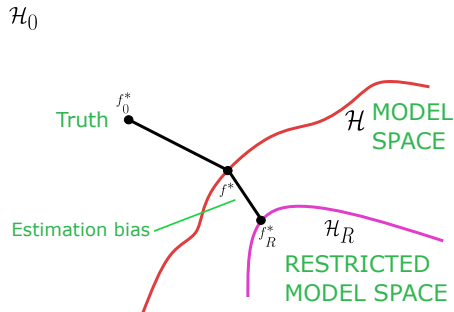
Two sources of error when an estimated model differs from the truth:

- ▶ **Bias**: how far the *average* estimate is from the truth
 - ▶ High bias = model class too restrictive to capture truth (underfit)
- ▶ **Variance**: how much the estimate *wiggles* across different training sets
 - ▶ High variance = model class so flexible it chases noise (overfit)

Regularization **increases bias** but **reduces variance**. Net effect on test error is often favorable.

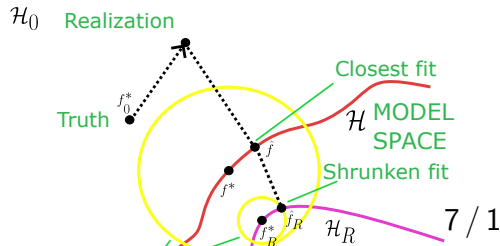
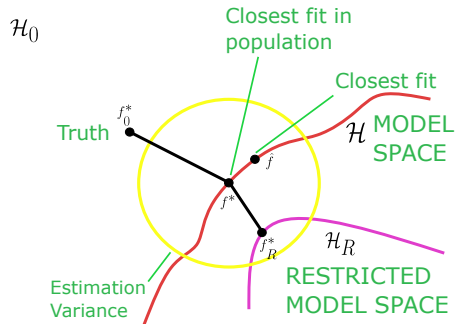
Visualizing the model spaces

- ▶ $\mathcal{H}_0$: “all” functions; ideal Bayes-optimal $f_0^* \in \mathcal{H}_0$
- ▶ $\mathcal{H}$: our chosen model class; best-in-class $f^* \in \mathcal{H}$
- ▶ $\mathcal{H}_R \subset \mathcal{H}$: regularized class
- ▶ $d(f_0^*, f^*) = \mathbf{model\ bias}$
- ▶ $d(f^*, f_R^*) = \mathbf{estimation\ bias}$ from regularization



Estimation variance shrinks under regularization

- ▶ Estimated model $\hat{f}$ depends on the random training sample
- ▶ **Estimation variance:** spread of $\hat{f}$ across resampled training sets
- ▶ Restricting to $\mathcal{H}_R$ *reduces* this spread
- ▶ Tradeoff: higher bias (we're constrained), lower variance



Ridge regression

Measure complexity as deviation from the origin. Shrink coefficients toward 0:

$$\begin{aligned}\hat{\boldsymbol{\theta}}_{\text{ridge}} &= \arg \min_{\boldsymbol{\theta}} \sum_{i=1}^n (y^{(i)} - \boldsymbol{\theta}^\top \mathbf{x}^{(i)})^2 + \lambda \sum_{j=1}^p \theta_j^2 \\ &= \arg \min_{\boldsymbol{\theta}} \|\mathbf{y} - \mathbf{X}\boldsymbol{\theta}\|_2^2 + \lambda \|\boldsymbol{\theta}\|_2^2\end{aligned}$$

Still has a **closed form**:

$$\hat{\boldsymbol{\theta}}_{\text{ridge}} = (\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I})^{-1} \mathbf{X}^\top \mathbf{y}$$

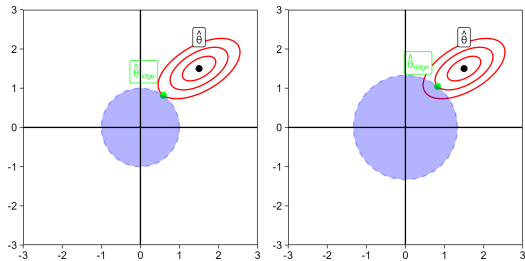
Name comes from adding positive entries along the diagonal “ridge” of $\mathbf{X}^\top \mathbf{X}$. Bonus: solves the singular- $\mathbf{X}^\top \mathbf{X}$ problem (multicollinearity).

Ridge as constrained optimization

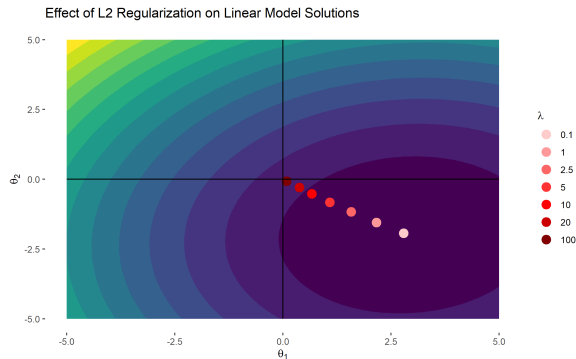
Equivalent constrained form (Lagrange duality):

$$\min_{\boldsymbol{\theta}} \sum_{i=1}^n (y^{(i)} - \boldsymbol{\theta}^\top \mathbf{x}^{(i)})^2 \quad \text{s.t.} \quad \|\boldsymbol{\theta}\|_2^2 \leq t$$

- ▶ Feasible region is an L_2 ball of radius $\sqrt{t}$
- ▶ Bijection: large $\lambda \Leftrightarrow$ small t
- ▶ Solution lies on the *boundary* of the ball (where it touches the smallest possible loss contour)

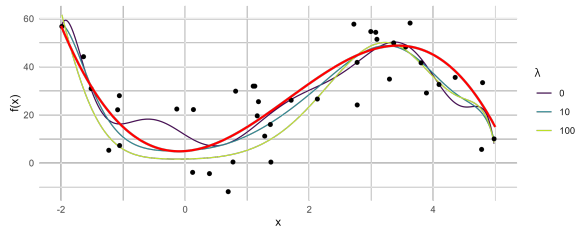


Ridge solution path



- ▶ Toy DGP: $y = 3x_1 - 2x_2 + \varepsilon$. True optimum $\theta^* = (3, -2)$.
- ▶ As $\lambda \uparrow$, $\hat{\theta}_{\text{ridge}}$ is pulled toward the origin
- ▶ Contours = unregularized objective; red points = ridge solutions at increasing λ
- ▶ Path is **smooth and non-sparse**

Why ridge helps: polynomial example



- ▶ True $f(x) = 5 + 2x + 10x^2 - 2x^3$ (red)
- ▶ Fit a degree-10 polynomial — overfits at $\lambda = 0$
- ▶ As λ grows: coefficients shrink, model gets smoother, generalization improves
- ▶ Lets us pick d *too large* on purpose and let λ control complexity (continuous, not discrete)

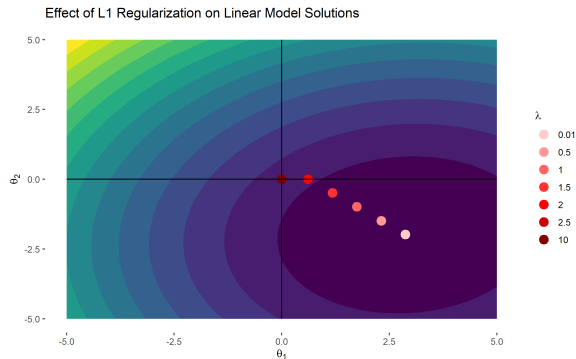
Lasso regression

Penalize the L_1 norm of the coefficients instead:

$$\hat{\boldsymbol{\theta}}_{\text{lasso}} = \arg \min_{\boldsymbol{\theta}} \|\mathbf{y} - \mathbf{X}\boldsymbol{\theta}\|_2^2 + \lambda \|\boldsymbol{\theta}\|_1 = \arg \min_{\boldsymbol{\theta}} \sum_{i=1}^n (y^{(i)} - \boldsymbol{\theta}^\top \mathbf{x}^{(i)})^2 + \lambda \sum_{j=1}^p |\theta_j|$$

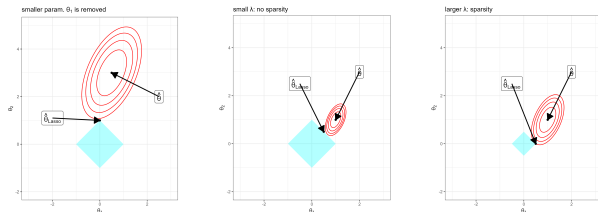
- ▶ “Least Absolute Shrinkage and Selection Operator”
- ▶ Still convex, but **not differentiable** at $\theta_j = 0 \Rightarrow$ no closed form
- ▶ Solve via coordinate descent, LARS, or proximal methods
- ▶ Key property: **sparsity** — many coefficients become *exactly* 0

Lasso solution path



- ▶ Same toy DGP ($y = 3x_1 - 2x_2 + \varepsilon$)
- ▶ Lasso path takes a *different route* than ridge
- ▶ Coefficients can become **exactly zero** for large enough λ
- ▶ Side effect: built-in **feature selection**

Lasso vs ridge: geometric intuition

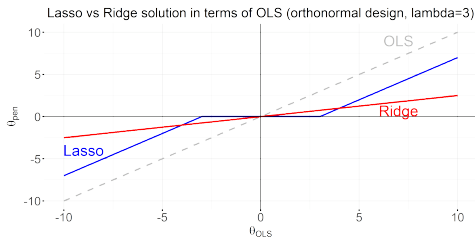


- ▶ Ridge constraint: L_2 ball (smooth)
- ▶ Lasso constraint: L_1 **diamond** with kinks on the axes
- ▶ Loss contours hit the diamond's **corners** first — corners lie on coordinate axes — so some θ_j become exactly 0
- ▶ Ridge's round ball is tangent to contours *between* the axes, so no sparsity

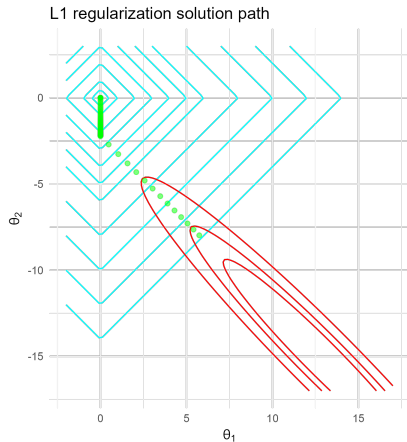
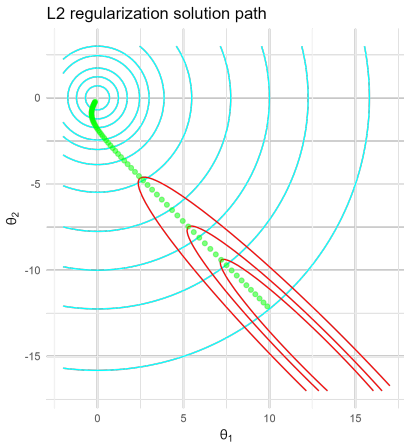
Soft thresholding (orthonormal $\mathbf{X}$)

In the special case $\mathbf{X}^\top \mathbf{X} = \mathbf{I}$, closed forms exist:

- ▶ **Lasso:** $\hat{\theta}_{\text{lasso}} = \text{sign}(\hat{\theta}_{\text{OLS}}) \cdot (|\hat{\theta}_{\text{OLS}}| - \lambda)_+$
 - ▶ “Soft thresholding”: small coefficients $\rightarrow 0$, large ones shrunk by λ
- ▶ **Ridge:** $\hat{\theta}_{\text{ridge}} = \hat{\theta}_{\text{OLS}} / (1 + \lambda)$
 - ▶ Uniform downscaling, no sparsity



Solution paths side-by-side



- **Ridge**: smooth, non-sparse path; all features stay in the model
- **Lasso**: piecewise-linear path; coefficients hit 0 one by one as λ grows

When to use which

Ridge (L_2):

- ▶ Many small-to-moderate effects expected
- ▶ Multicollinearity present (helps numerically)
- ▶ Closed-form fit is desirable (cheap)
- ▶ You don't need exact zeros

Lasso (L_1):

- ▶ You suspect only a few features matter (sparse truth)
- ▶ You want automatic feature selection
- ▶ Interpretability matters

Elastic Net = $\alpha\|\boldsymbol{\theta}\|_1 + (1 - \alpha)\|\boldsymbol{\theta}\|_2^2$: blends both. Good default when in doubt.

Wrap-up

1. Regularization adds an inductive bias toward “simple” models
2. Common pattern: $\mathcal{R}_{\text{reg}}(\boldsymbol{\theta}) = \mathcal{R}_{\text{emp}}(\boldsymbol{\theta}) + \lambda J(\boldsymbol{\theta})$
3. Ridge (L_2): smooth shrinkage, closed-form, no sparsity
4. Lasso (L_1): kinked, sparse — does feature selection for free
5. λ controls the bias-variance tradeoff; pick it via CV (L05)

Next: L04 — how do we even *measure* “test error” vs “training error”? When is a model overfitting? How do we evaluate regression metrics honestly?